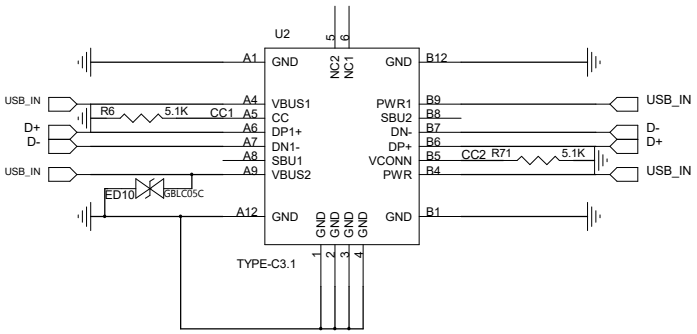
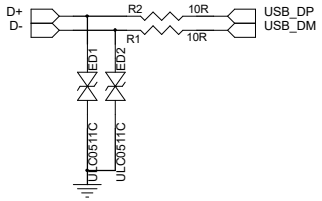
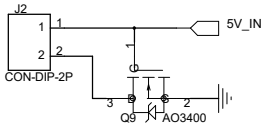
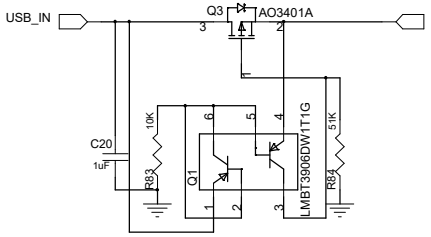
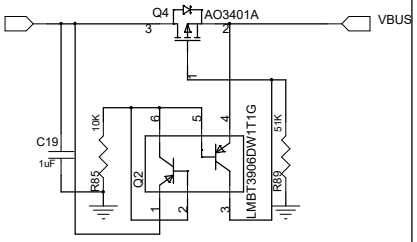
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



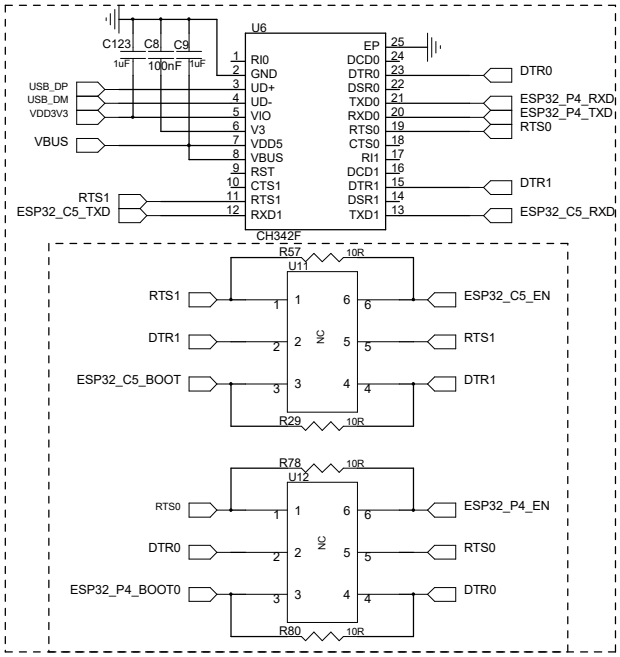


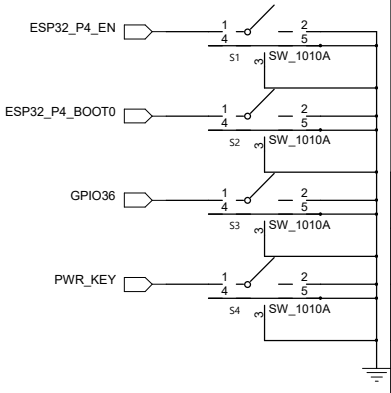






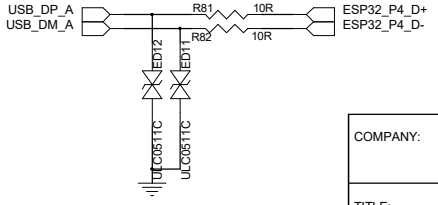
POWER IN

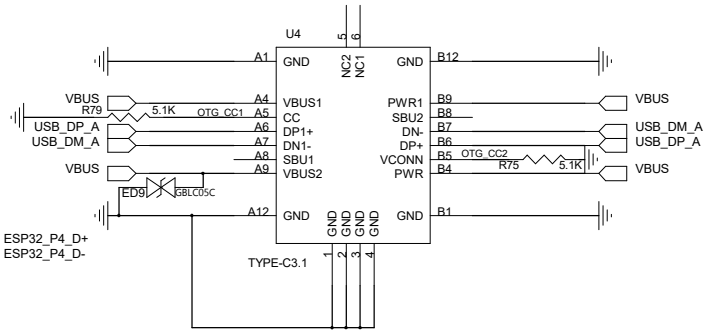




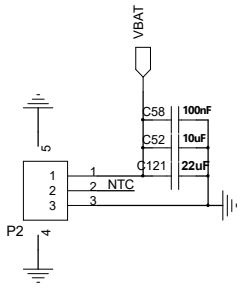
USB TO UART

USB OTG

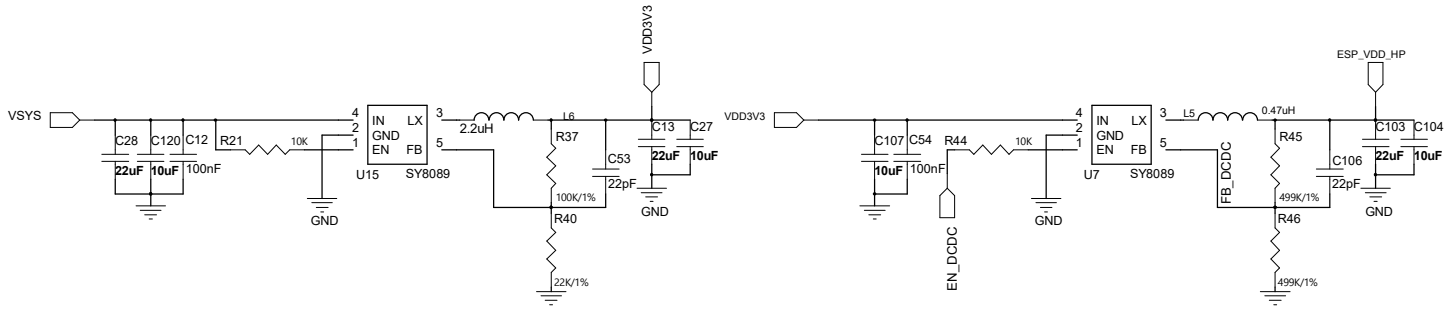




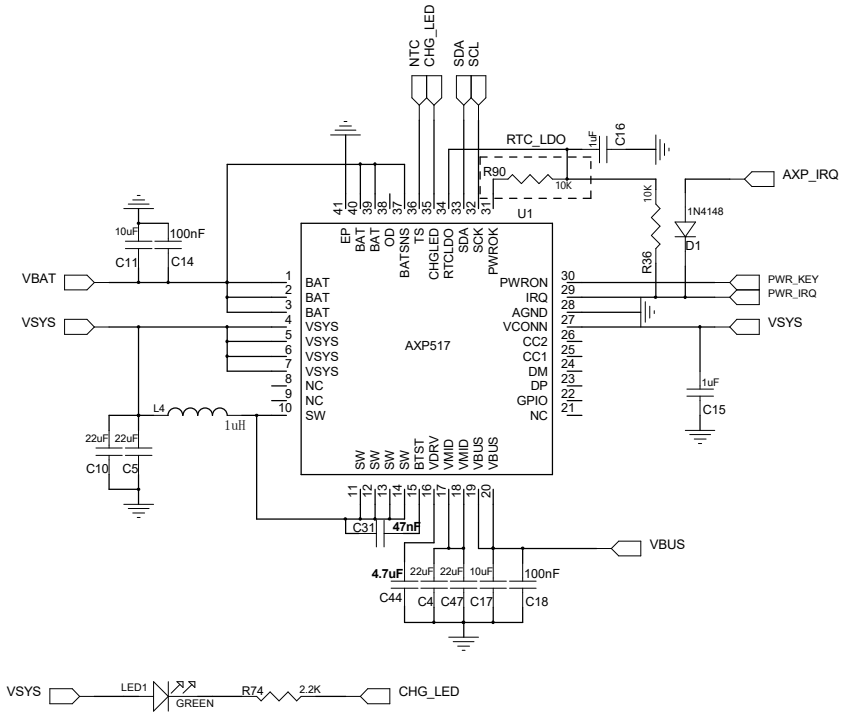
DRAWN:	DATED:	COMPANY:			
		TITLE:			
CHECKED:	DATED:	CODE:	SIZE: B	DRAWING NO:	REV:
QUALITY CONTROL:	DATED:				
RELEASED:	DATED:	SCALE:		SHEET: 1 OF 8	



Battery Interface



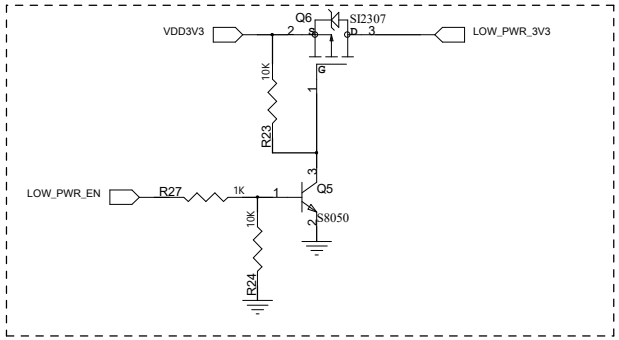
MAIN POWER

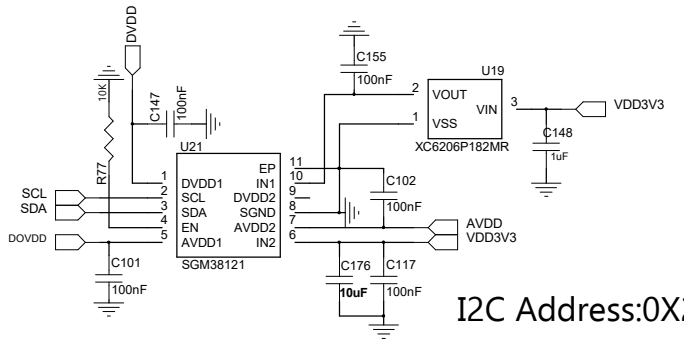


SDA-------IO3
SCL-------IO2
AXP_IRQ-------IO4

I2C Address:0X34

LOW POWER

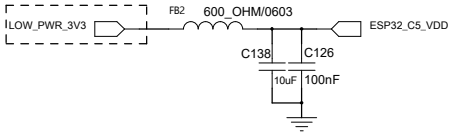


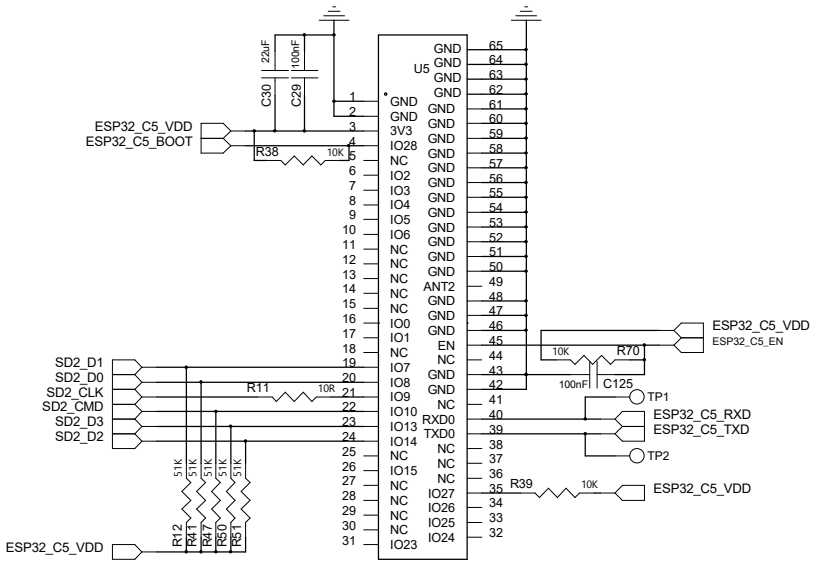


I2C Address:0X28

Title		
Size B	Document Number POWER-INPUT	Rev V1.6
Date: Thursday, September 17, 2020	Sheet 2 of 8	

Core

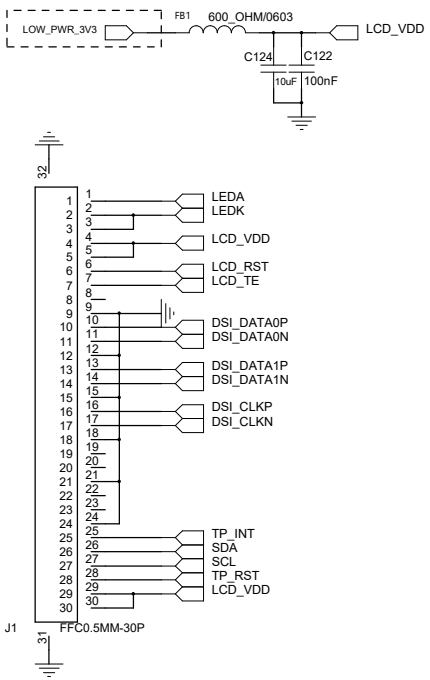




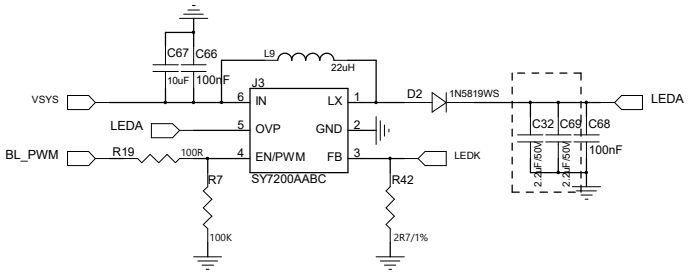
C5_D0-------IO14
C5_D1-------IO15
C5_D2-------IO16
C5_D3-------IO17
C5_CLK-------IO18
C5_CMD-------IO19
C5_EN-------IO54

Title		
Size B	Document Number ESP32	Rev V1.6
Date: Wednesday, December 16, 2020	Sheet 3 of 8	

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



SDA-------IO3
SCL-------IO2
LCD_TE-------IO33
TP_INT-------IO6
BL_PWM-------IO34



I2C ADDRESS:0x68

DRAWN:	DATED:
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

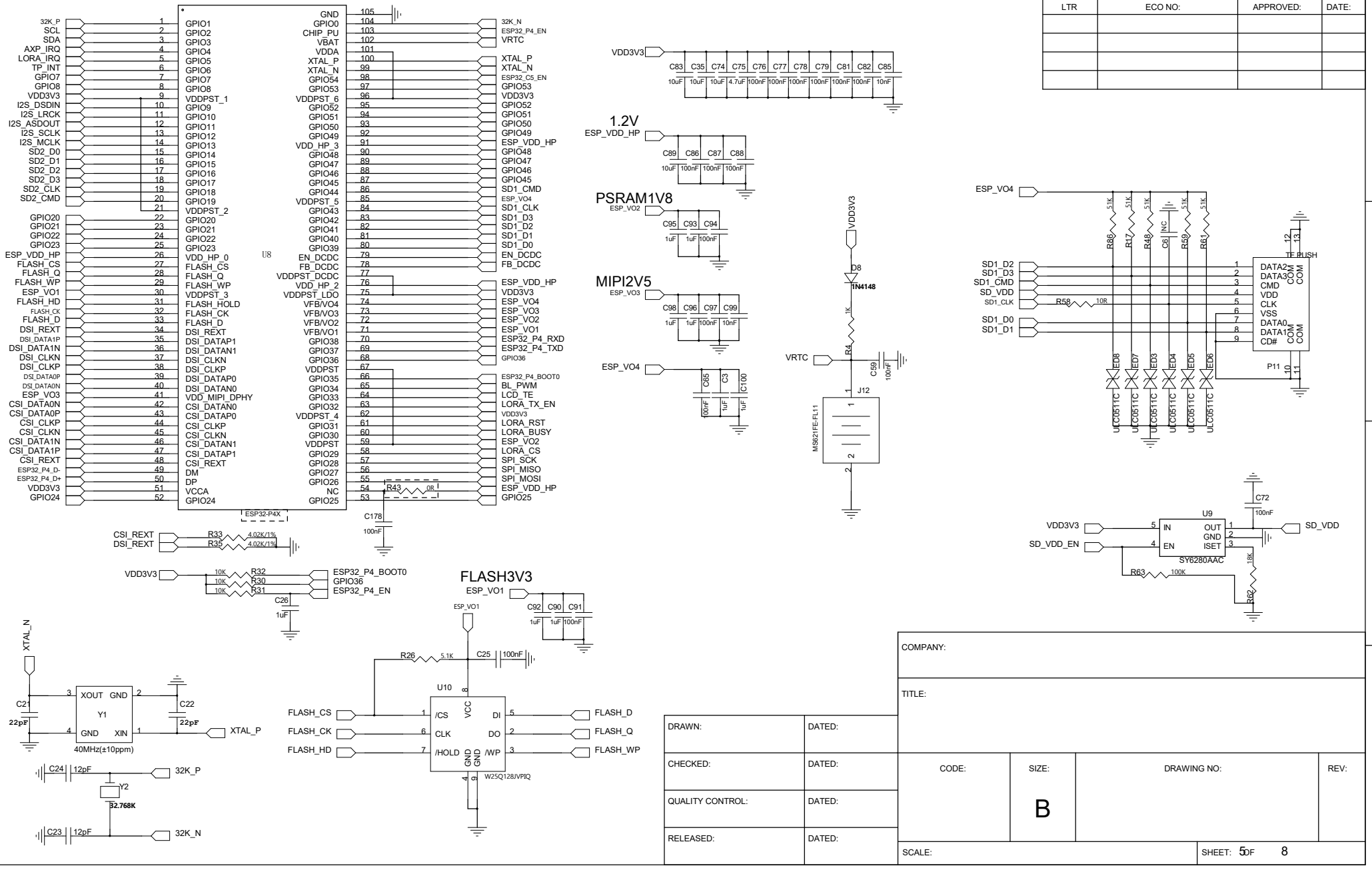
COMPANY:

TITLE:

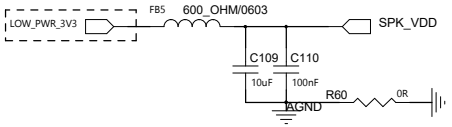
CODE:	SIZE:	DRAWING NO:	REV:
	B		

SCALE:

SHEET: 4 OF 8


REVISION RECORD
LTR
ECO NO:
APPROVED:
DATE:
U8
ESP32-P4X
GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 VDDPST_1 GPIO9 GPIO10 GPIO11 GPIO12 GPIO13 GPIO14 GPIO15 GPIO16 GPIO17 GPIO18 GPIO19 VDDPST_2 GPIO20 GPIO21 GPIO22 GPIO23 VDD_HP_0 FLASH_CS FLASH_Q FLASH_WP FLASH_HOLD FLASH_CK FLASH_D FLASH_REXT DSI_DATAP1 DSI_DATAN1 DSI_CLKN DSI_CLKP DSI_DATAP0 DSI_DATAN0 VDD_MIPI_DPHY CSI_DATAN0 CSI_DATAP0 CSI_CLKP CSI_CLKN CSI_DATAN1 CSI_DATAP1 CSI_REXT DM DP VCCA GPIO24
GND GPIO0 CHIP_PU VBAT VDDA XTAL_P XTAL_N GPIO54 GPIO53 VDDPST_6 GPIO52 GPIO51 GPIO50 GPIO49 VDD_HP_3 GPIO48 GPIO47 GPIO46 GPIO45 GPIO44 VDDPST_5 GPIO43 GPIO42 GPIO41 GPIO40 GPIO39 EN_DCDC FB_DCDC VDDPST_DCDC VDD_HP_2 VDDPST_LDO VFB/VO4 VFB/VO3 VFB/VO2 VFB/VO1 GPIO38 GPIO37 GPIO36 VDDPST GPIO35 GPIO34 GPIO33 GPIO32 VDDPST_4 GPIO31 GPIO30 VDDPST GPIO29 GPIO28 GPIO27 GPIO26 NC GPIO25
32K_P SCL SDA AXP_IRQ LORA_IRQ TP_INT GPIO7 GPIO8 VDD3V3 I2S_DSDIN I2S_LRCK I2S_ASDOUT I2S_SCLK I2S_MCLK SD2_D0 SD2_D1 SD2_D2 SD2_D3 SD2_CLK SD2_CMD
ESP_VDD_HP FLASH_CS FLASH_Q FLASH_WP ESP_VO1 FLASH_HD FLASH_CK FLASH_D DSI_REXT DSI_DATA1P DSI_DATA1N DSI_CLKN DSI_CLKP DSI_DATA0P DSI_DATA0N ESP_VO3 CSI_DATA0N CSI_DATA0P CSI_CLKP CSI_CLKN CSI_DATA1N CSI_DATA1P CSI_REXT ESP32_P4_D- ESP32_P4_D+ VDD3V3 GPIO24
32K_N ESP32_P4_EN VRTC XTAL_P XTAL_N ESP32_C5_EN GPIO53 VDD3V3 GPIO52 GPIO51 GPIO50 GPIO49 ESP_VDD_HP GPIO48 GPIO47 GPIO46 GPIO45 SD1_CMD ESP_VO4 SD1_CLK SD1_D3 SD1_D2 SD1_D1 SD1_D0 EN_DCDC FB_DCDC ESP_VDD_HP VDD3V3 ESP_VO4 ESP_VO3 ESP_VO2 ESP_VO1 ESP32_P4_RXD ESP32_P4_TXD GPIO36 ESP32_P4_BOOT0 BL_PWM LCD_TE LORA_TX_EN VDD3V3 LORA_RST LORA_BUSY ESP_VO2 LORA_CS SPI_SCK SPI_MISO SPI_MOSI ESP_VDD_HP GPIO25
R43 0R
C178 100nF
CSI_REXT R33 4.02K/1%
DSI_REXT R35 4.02K/1%
VDD3V3 R32 R30 R31 10K
ESP32_P4_BOOT0
GPIO36
ESP32_P4_EN
C26 1uF
XTAL_N
C21 22pF
C22 22pF
Y1
XOUT GND XIN
40MHz(±10ppm)
XTAL_P
C24 12pF
Y2
32.768K
C23 12pF
32K_P
32K_N
FLASH3V3
ESP_VO1
C92 1uF
C90 1uF
C91 100nF
R26 5.1K
C25 100nF
U10
W25Q128JVPIQ
FLASH_CS FLASH_CK FLASH_HD
/CS /HOLD CLK VCC GND GND DI DO /WP
FLASH_D FLASH_Q FLASH_WP
VDD3V3
C83 10uF C35 10uF C74 10uF C75 4.7uF C76 100nF C77 100nF C78 100nF C79 100nF C81 100nF C82 100nF C85 10nF
1.2V
ESP_VDD_HP
C89 10uF C86 100nF C87 100nF C88 100nF
PSRAM1V8
ESP_VO2
C95 1uF C93 1uF C94 100nF
MIPI2V5
ESP_VO3
C98 1uF C96 1uF C97 100nF C99 10nF
ESP_VO4
C65 100nF C3 1uF C100 1uF
VDD3V3
D8 1N4148
1K R4
VRTC
C59 100nF
J12
M5621FE-FL11
ESP_VO4
R86 R17 R48 C6 INC R59 R61 5.1K
SD1_D2 SD1_D3 SD1_CMD SD_VDD SD1_CLK R58 10R
SD1_D0 SD1_D1
DATA2 DATA3 CMD VDD CLK VSS DATA0 DATA1 CD#
TF PUSH
P11
ED8 ED7 ED3 ED4 ED5 ED6 UC0511C
C72 100nF
U9
SY6280AAC
VDD3V3 SD_VDD_EN
IN EN OUT GND ISET
SD_VDD
R63 100K
R62 18K
COMPANY:
TITLE:
DRAWN: DATED:
CHECKED: DATED:
QUALITY CONTROL: DATED:
RELEASED: DATED:
CODE:
SIZE: B
DRAWING NO:
REV:
SCALE:
SHEET: 5 OF 8

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



SDA-------IO3

SCL-------IO2

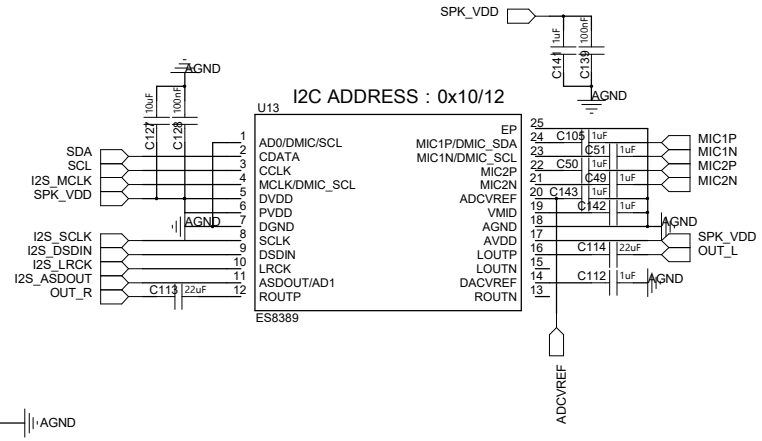
I2S_MCLK------IO13

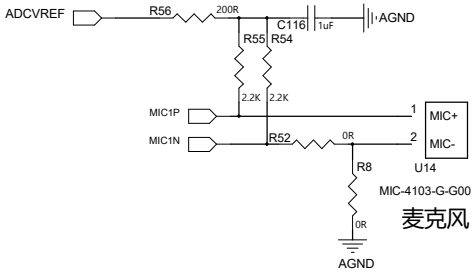
I2S_SCLK------IO12

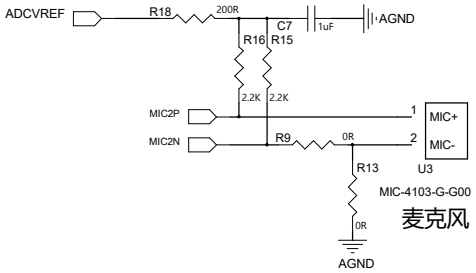
I2S_ASDOUT------IO11

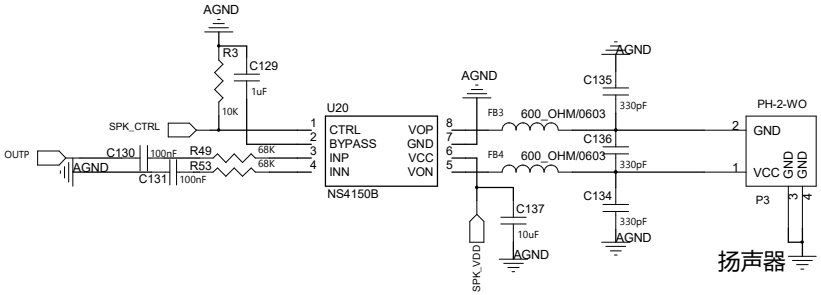
I2S_LRCK------IO10

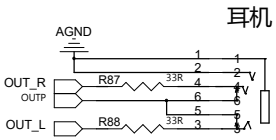
I2S_DSDIN------IO9











DRAWN:	DATED:
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

COMPANY:

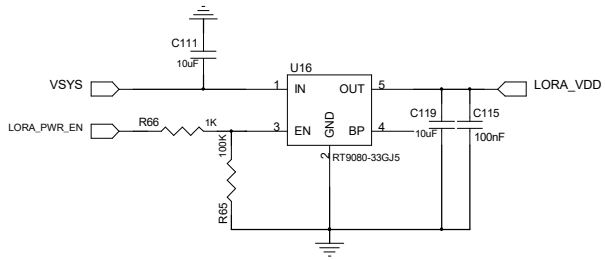
TITLE:

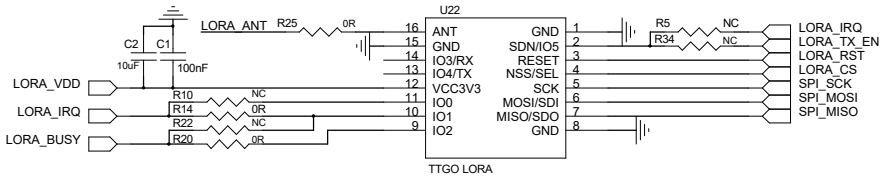
CODE:	SIZE:	DRAWING NO:	REV:
	B		

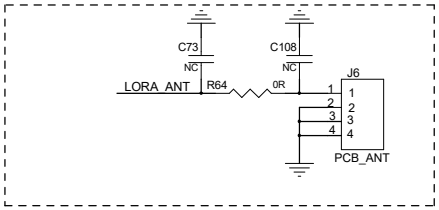
SCALE:

SHEET: 6 OF 8

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:







LORA_IRQ------IO5
LORA_BUSY------IO30
SPI_MISO------IO27
SPI_MOSI------IO26
SPI_SCK------IO28
LORA_CS------IO29
LORA_RST------IO31
LORA_TX_EN------IO32

DRAWN:	DATED:
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

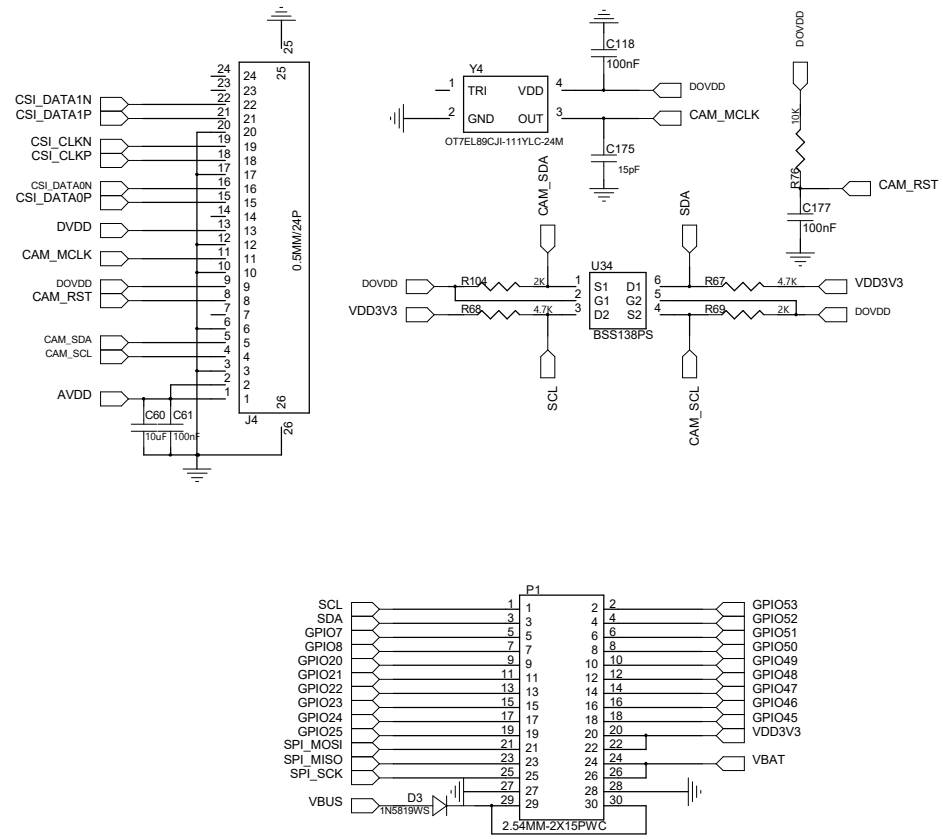
COMPANY:

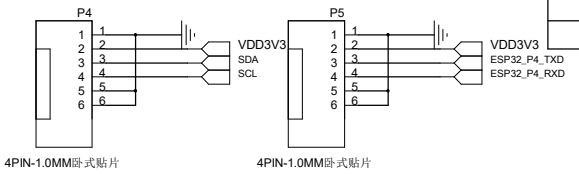
TITLE:

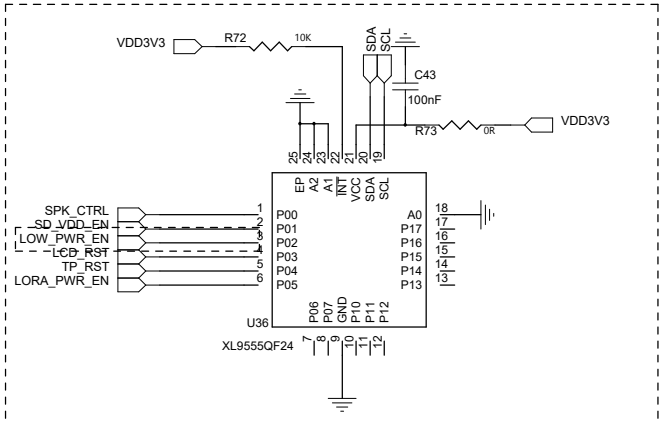
CODE:	SIZE:	DRAWING NO:	REV:
	B		

SCALE:

SHEET: 7 OF 8







REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

DRAWN:	DATED:
CHECKED:	DATED:
QUALITY CONTROL:	DATED:
RELEASED:	DATED:

COMPANY:

TITLE:

CODE:	SIZE:	DRAWING NO:	REV:
	B		

SCALE:

SHEET: 8 OF 8